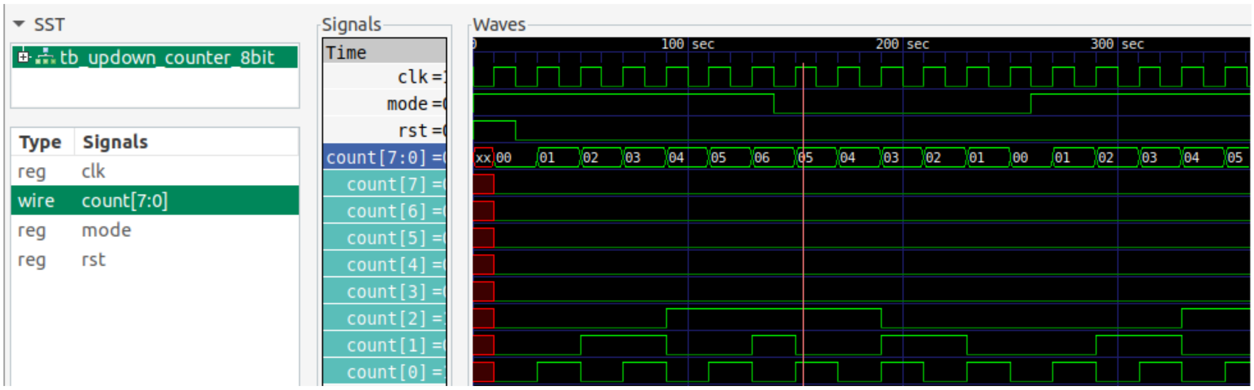
Circuit Simulation Lab:20
Design of Counters

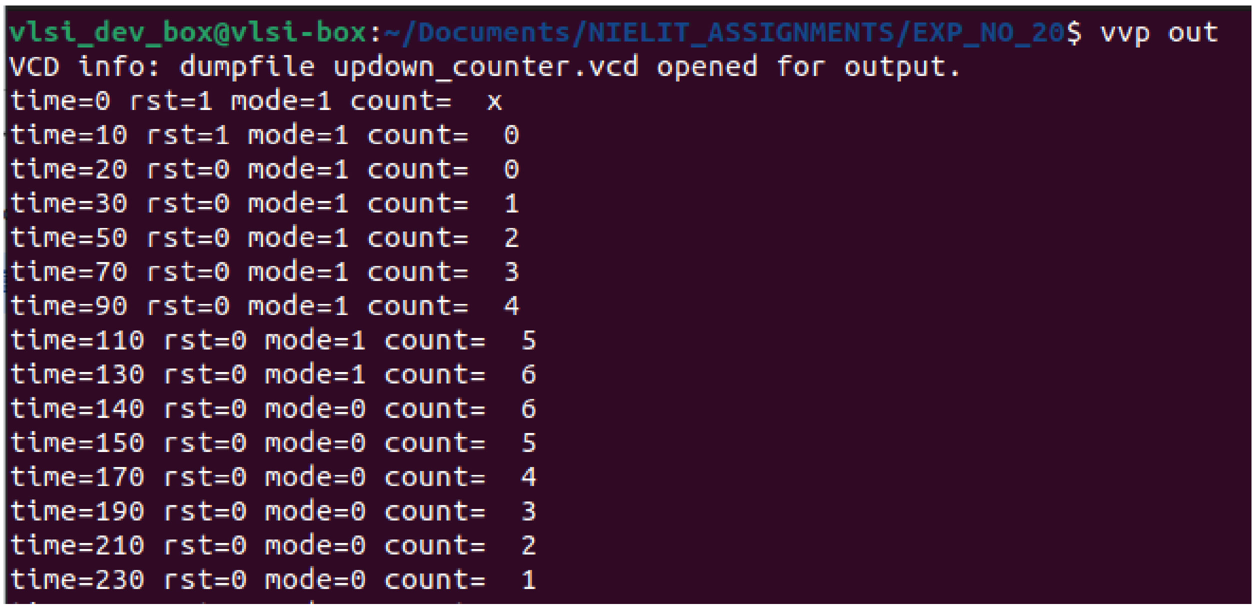
The purpose of this experiment is to introduce the design of Synchronous sequential circuits, in this case we are taking an 8 bit up/down counter

CODE:
```
module updown_counter_8bit(
input clk,
input rst,
input mode,
output reg [7:0] count
);

always @(posedge clk)
begin

if(rst)
   count <= 8'd0;

else if(mode)
   count <= count + 1'b1;

else
   count <= count - 1'b1;

end

Endmodule
```

TESTBENCH
```
module tb_updown_counter_8bit;

reg clk;
reg rst;
reg mode;

wire [7:0] count;

updown_counter_8bit dut(
```

```
.clk(clk),
.rst(rst),
.mode(mode),
.count(count)
);

always #10 clk = ~clk;

initial begin

$dumpfile("updown_counter.vcd");
$dumpvars(0,tb_updown_counter_8bit);

clk  = 0;
rst  = 1;
mode = 1;

#20;
rst = 0;

// Count Up
mode = 1;
#120;

// Count Down
mode = 0;
#120;

// Count Up Again
mode = 1;
#120;

$finish;

end

initial begin
$monitor("time=%0t rst=%b mode=%b count=%d",
        $time,rst,mode,count);
end

Endmodule
```





```
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_20$ vvp out
VCD info: dumpfile updown_counter.vcd opened for output.
time=0 rst=1 mode=1 count=  x
time=10 rst=1 mode=1 count=  0
time=20 rst=0 mode=1 count=  0
time=30 rst=0 mode=1 count=  1
time=50 rst=0 mode=1 count=  2
time=70 rst=0 mode=1 count=  3
time=90 rst=0 mode=1 count=  4
time=110 rst=0 mode=1 count=  5
time=130 rst=0 mode=1 count=  6
time=140 rst=0 mode=0 count=  6
time=150 rst=0 mode=0 count=  5
time=170 rst=0 mode=0 count=  4
time=190 rst=0 mode=0 count=  3
time=210 rst=0 mode=0 count=  2
time=230 rst=0 mode=0 count=  1
```